

Core Lab – III : Data Structures Lab

Course Code	Course Name	Category	Hours /Week	Credits
25BCA34P	Data Structures using Lab	Core Lab - III	4	2

S.No	List of Programs	
1	Implementation of Array Operations.	
2	Implementation of Stack using arrays.	
3	Implementation of Queue using arrays.	
4	Implementation of Singly Linked List	
5	Conversion of Infix to Postfix Expression.	
6	Evaluation of Postfix Expression.	
7	Polynomial Addition.	
8	Implementation of Binary Tree and Binary tree traversal techniques.	
9	Searching Techniques: Linear search and Binary search	
10	Sorting Techniques: Insertion Sort, Quick Sort, Radix Sort, Heap Sort.	
Total Hours		60

Text Books

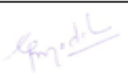
1. Balaguruswamy E(2013), Object Oriented Programming Through C++, 6th Edition.
2. Varsha H. Patil, Data Structures using C++, Oxford.

Reference Books

1. Ellis Horowitz, Sartaj Sahani and Dinesh Mehta (2008), Fundamentals of Data Structures in C++, 2nd Edition, University Press.
2. Mark Allen Weiss, Data Structures and Algorithm Analysis in C, 2nd Edition, Pearson Education Asia.

Web Resources (Swayam / NPTEL)

1. https://onlinecourses.swayam2.ac.in/nou24_cs06/preview
2. https://onlinecourses.swayam2.ac.in/cec19_cs04/preview
3. https://onlinecourses.swayam2.ac.in/aic20_sp06/preview

Course	Name	Signature
Prepared By	Dr.G.Priyadarshini, Faculty, Department of Computer Applications	
Vetted By	Dr.P.Vijayakumar, HoD, Department of Computer Applications	
Validated By	BoS Chairman	